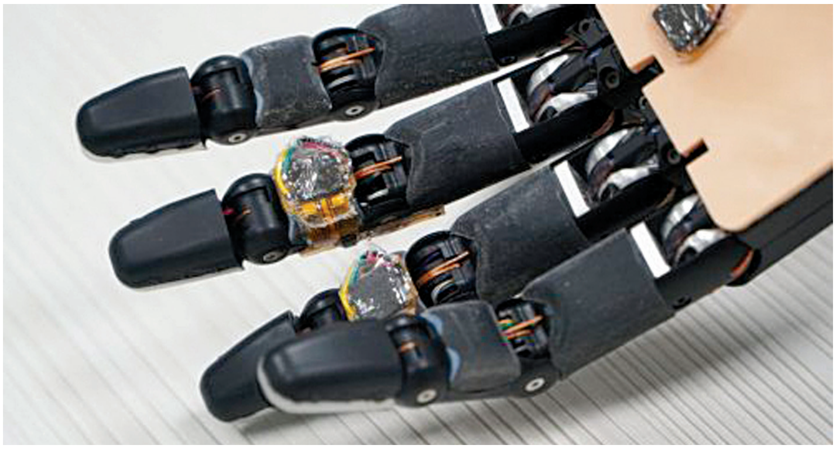
tiny electrodes underneath it. By applying pressure, the metal particles came closer to change the electrical properties. These changes were then detected by the electrodes to give commands to the robot. This feature makes robots more intelligent and interactive as it enables the robotic hand to detect not only the amount but also the direction of applied force.



Robotic arm (Credit: National University Singapore's Materials Sciences and Engineering lab)

Biofuel cells produce electricity in eco-friendly way

Button or coin cells powering portable devices are difficult to recycle, toxic and dangerous to the environment. This calls for the development of safe and ecological alternatives. A company called BeFC Bioenzymatic Fuel Cells has developed a paper based, ultra-thin, flexible, and miniature bio-enzymatic fuel cell system that uses biological catalysts instead of chemical or expensive noble metal catalysts. It works by converting natural substrates such as glucose and oxygen into electricity. This metal- and plastic-free technology provides a sustainable and environmentally friendly method of generating energy.

Small-size bioenymatic fuel cell (Credit: www.befc.global)

Radar beams for networking and localising objects

The Fraunhofer Institute for Reliability and Microintegration IZM in its project called OmniConnect has used a secondary radar to detect the motion and position of objects inside rooms. Conventional radar detects objects and their movement but does not provide any other data. A secondary radar combines radar beams with tags that are attached to objects. The radar is free from any conflicts with mobile radio networks, Wi-Fi, or Bluetooth. Also, the emitted radiation is completely harmless to human beings. The system is an ideal solution for networking any objects or everyday items with one another or integrating them into a home network. And because the passive tags do not need a separate power supply, there's no inconvenience of having to replace batteries.

The passive tags can be unobtrusively integrated into everyday objects such as items of clothing. Radar modules on the ceiling detect the position and the movement of the tags (Credit: www.fraunhofer.de/en/press)